00287

Sri Lanka Institute of Information Technology

B.Sc. Honours Degree in Information Technology

Specialized in Information Technology

Final Examination
Year 2, Semester 2 (2024)

IT2020 – Software Engineering

Duration: 3 Hours

November 2024

Instructions to Candidates:

- This paper has 5 questions.
- Answer all questions in the booklet given.
- The total mark for the paper is 100.
- This paper contains 8 pages, including the cover page.
- Electronic devices capable of storing and retrieving text, including calculators and mobile phones are not allowed.

Question 1 (25 marks)

Answer the questions based on the following "Schedule Appointment" use case scenario in Table 1.

Note: You may use suitable boundary, control, and entity classes.

Table 1: "Ordering Food" Use Case Scenario

Usecase Name: Ordering Food
Actor: Customer
Main Flow of Events:
1. The customer enters his/her username and password into the LoginUI to log into the system.
2. The system validates the customer's credentials.
3. If the login is successful, the system displays a success message.
4. A customer can order multiple dishes.
5. For each dish, the customer navigates to the MenuUI, where the system displays the restaurant's menu options.
6. The customer selects the desired dishes from the menu and submits the order to the system with dish_ID.
7. The system checks the availability of the selected dishes in dishes_DB and displays the availability of the dish to the customer.
8. If the dish is available, the customer confirms the order.
9. Then the system place the order via order_DB which notifies the kitchen to start preparing the order.
10. Simultaneously the system generates the invoice and display to the customer.
Extension:
3a. If the login is unsuccessful, the system prompts the customer to reenter their login details.

a) Draw a sequence diagram to represent step 1 to 3. (05 marks)

b) Draw another sequence diagram to represent the whole scenario in Table 1 by referencing the sequence diagram in part (a). (20 marks)

Question 2 (20 marks)

Draw a **state machine diagram** that illustrates the states and transitions for the Air Conditioner based on the scenario given below.

By default, the Air conditioner is in Off mode. When the Air Conditioner is turned on, it enters the CoolMode state, setting its target temperature to 24°C and displaying a blue light on the indicator panel. It gradually cools the room until it reaches the target temperature. While in CoolMode, the user can switch to FanMode by pressing the fan mode button. In FanMode, the target temperature is disabled, the cooling compressor is turned off, and the indicator light changes to green. The fan speed gradually increases to circulate air.

User can switch back to CoolMode from FanMode by pressing the cool mode button, where the indicator light turns blue again, and the target temperature is restored to 24°C. While in the Off mode, the user can also switch to HeatMode by pressing the heat mode button. In HeatMode, the target temperature is set to 28°C, and the indicator light changes to red. The AC gradually warms the room until it reaches the target temperature.

The Air Conditioner can be turned off in any state by switching off the power. While in the CoolMode, HeatMode or FanMode, if a system malfunction, such as a refrigerant leak, is detected, the AC transitions to SafetyMode within 30 seconds. SafetyMode consists of two sub-states: Diagnostics and SystemRecovery. The AC first enters Diagnostics, where it performs checks to identify the problem. After diagnostics, it proceeds to SystemRecovery, where it attempts to resolve the issue.

The AC can be turned off in SafetyMode or it can automatically move to off state after 30 minutes if not recovered. After successful recovery in SafeMode, the AC can return to CoolMode.

Question 3 (20 marks)

Part A

Answer the questions given below using the partial code segment in Figure 1.

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        int choice;

        System.out.print("Enter your choice (Enter 0 to exit): ");
        choice = scanner.nextInt();

        while (choice != 0) {
            switch (choice) {
                case 1:
                    System.out.println("Green");
                    break;
                case 2:
                    System.out.println("Orange");
                    break;
                case 3:
                    System.out.println("Blue");
                    break;
                default:
                    System.out.println("Invalid option");
            }

            System.out.print("Enter your choice: ");
            choice = scanner.nextInt();
        }

        scanner.close();
    }
}
```

Figure 1: Partial coding segment

a) Draw a control flow graph for this code segment (Note: Showing correct branches of the code will be enough) (04 marks)

b) Consider the above code (Figure 1) and calculate the branch coverage percentage when input 1,2,0 values in a sequence. (02 marks)

c) Consider the above code (Figure 1) and calculate the branch coverage percentage when input 1,2,3,6,0 values in a sequence. (02 marks)

Part B

Answer the questions given below using the partial code segment in Figure 2.

```
public class GradeCalculator {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);

        System.out.print("Enter student score: ");
        int score = scanner.nextInt();

        System.out.print("Enter student attendance percentage: ");
        int attendance = scanner.nextInt();

        if (attendance < 75) {
            System.out.println("Fail (due to low attendance)");
        }
        else {
            if (score >= 90) {
                System.out.println("Grade: A");
            }
            else if (score >= 80) {
                System.out.println("Grade: B");
            }
            else if (score >= 70) {
                System.out.println("Grade: C");
            }
            else {
                System.out.println("Fail (due to low score)");
            }
        }

        scanner.close();
    }
}
```

Figure 2: Code segment for Student's grade based on their score

a) Identify the number of test cases needed for full statement coverage for the above code in Figure 2. Calculate statement coverage for each test case. (10 marks)

b) Explain the main difference between "Black box testing" and "White box testing" in software testing. (02 marks)

Question 4 (20 marks)

use the proper question number and subsection formatting

a)

You have given the task of designing a feature for an online shopping platform that allows customers to purchase various items. Each item can have optional add-ons such as gift wrapping, express delivery, and a personalized message. The platform needs to support adding multiple optional add-ons to any item dynamically. Also, each add-on has a specific cost, which should be reflected in the final price of the item. The system should be flexible enough to allow future add-ons to be added without requiring changes to the existing classes.

i) Recommend the most suitable design pattern for the given scenario and justify your recommendation. (02 marks)

ii) Draw the class structure of the design pattern that you identified above in part i) with appropriate classes and methods for the above scenario. (05 marks)

b)

Assume you are designing a stock market monitoring system for investors. The system tracks various stocks and notifies subscribers (investors) whenever there are significant changes in the stock prices. Investors may want to receive updates through different channels, such as email, SMS, or a mobile app notification. Additionally, investors may subscribe or unsubscribe from notifications at any time.

i) Recommend the most suitable design pattern for the given scenario above and justify your recommendation. (02 marks)

ii) Draw the class structure of the design pattern that you identified above in part i) with appropriate classes and methods for the above scenario. (05 marks)

c) Consider the code segment in Figure 3 and answer the questions provided at the end.

```
class DVDPlayer {
    public void on() { System.out.println("DVD Player on"); }
    public void play(String movie) { System.out.println("Playing " + movie); }
    public void off() { System.out.println("DVD Player off"); }
}

class Projector {
    public void on() { System.out.println("Projector on"); }
    public void wideScreenMode() { System.out.println("Widescreen mode"); }
    public void off() { System.out.println("Projector off"); }
}

class HomeTheater {
    private DVDPlayer dvd;
    private Projector projector;

    public HomeTheater(DVDPlayer dvd, Projector projector) {
        this.dvd = dvd;
        this.projector = projector;
    }

    public void watchMovie(String movie) {
        dvd.on();
        projector.on();
        projector.wideScreenMode();
        dvd.play(movie);
    }

    public void endMovie() {
        dvd.off();
        projector.off();
    }
}


public class Test {
    public static void main(String[] args) {
        DVDPlayer dvd = new DVDPlayer();
        Projector projector = new Projector();
        HomeTheater homeTheater = new HomeTheater(dvd, projector);

        homeTheater.watchMovie("Inception");
        homeTheater.endMovie();
    }
}
```

Figure 3: "Home Theater System" Code segment

i) Identify the design pattern used in the code in Figure 3. (01 mark)

ii) Identify the main purpose of using the design pattern you have mentioned above in part i) for this solution. (01 mark)

iii) Draw the class structure of the design pattern that you identified in part i) with appropriate methods to represent the above scenario. (04 marks)

Question 5 (15 marks)

a) Name the the key components of configuration management in software engineering. (02 marks)

b) Identify the key features of version control systems that support project management and development. (03 marks)

c) list the advantages of using Software as a Service (SaaS) compared to traditional packaged software. (02 marks)

d) Identify the three types of internal behaviors in a state, and how do they function. (04 marks)

e) Identify the key components represented in a communication diagram. (02 marks)

f) Explain the main difference between the sequence diagrams and communication diagrams. (02 marks)

---------- END OF THE QUESTION PAPER ----------